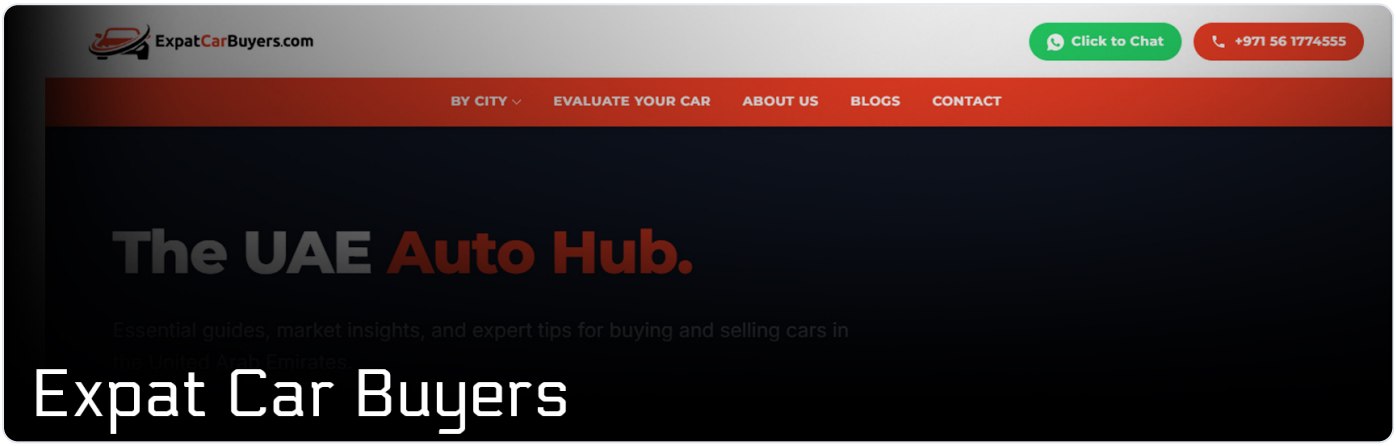
FINTECH & AUTOMOTIVE E-COMMERCE

ExpatCarBuyers: High-Performance UAE Car Valuation Platform

Live Production Platform: https://www.expatcarbuyers.com | Instant valuation, free inspection, cash in 30 minutes



1. Executive Summary & Overview

Sell any car in the UAE with instant valuation, free professional inspection, and cash in hand within 30 minutes. We engineered the complete full-stack platform for ExpatCarBuyers, operating across Dubai, Abu Dhabi, and Sharjah.

The system replaced a fragmented legacy system with a modern, performant architecture: a **Laravel 11 API** backend on **Neon PostgreSQL** with a **Filament admin panel**, and a **Next.js 15 (App Router)** frontend with **Meilisearch-powered instant search**. The valuation flow went from a brittle multi-page form to a single-page, offline-ready wizard that preloads the entire 50k-variant catalog in the background and responds in <50 ms per step.

2. THE ENGINEERING CHALLENGE

Making a **49,746-variant car catalog** feel instantaneous on constrained resources. Standard relational queries caused severe bottlenecks, while competitors suffered from slow multi-step forms.

3. THE ARCHITECTURAL SOLUTION

Versioned cache keys, JOIN-based queries eliminating N+1 lookups, a "Smart Warm" preload endpoint covering ~95% of sessions instantly, and zero-downtime Neon PostgreSQL database migration.

4. Core Architectural Features

Multi-Step Valuation Wizard
Year → Make → Model → Variant in a single-page flow with Zod validation, Framer Motion animations, and local caching.

Smart Warm Catalog Preload
Returns years + latest-year makes + top-20 models in one cached response (~95% session coverage) with offline sync.

Meilisearch Instant Search
Client-side typeahead across 49k+ variants with automatic PostgreSQL ILIKE fallback.

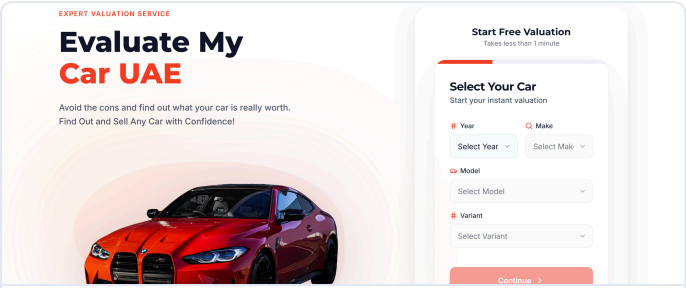
Filament v4 Admin Panel

Full operational management for Bookings, Makes, Models, Variants, Branches, and Staff with role-based access.

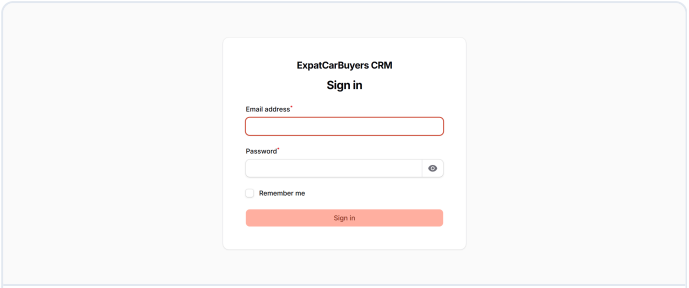
Inspection Booking System
7-day rolling calendar with 30-min slots for Office and Home inspection dispatch.

Queued Notification Pipelines
Instant automated customer SMS (SMS Global) and transactional emails via Laravel Horizon queues.

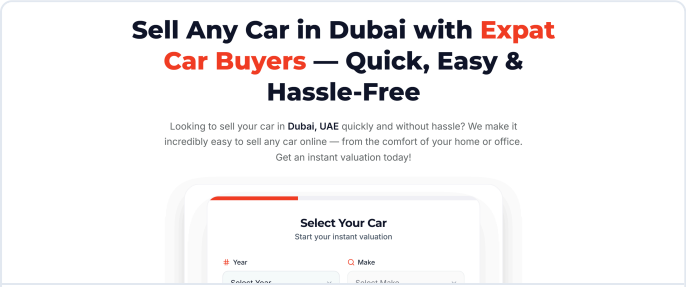
5. User Interface & Administration Gallery



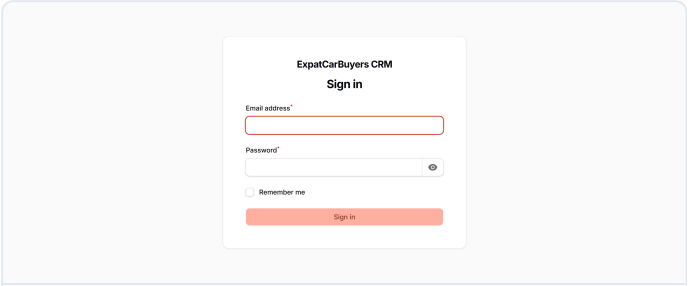
Single-Page Rapid Car Valuation Engine



Filament Admin Executive Dashboard



Public SEO-Optimized Landing Page



Branch & Operational Location Hub

6. Complete Technology Stack

Layer	Technology Implementation
Backend API	Laravel 11 (PHP 8.2), Laravel Horizon, Spatie MediaLibrary
Frontend Application	Next.js 15 (App Router, React 18, TypeScript), Tailwind CSS
Database	PostgreSQL 18 on Neon (Serverless, eu-central-1)
Admin Infrastructure	Filament v4 (Role-based access control, real-time metrics widgets)
Search & Cache	Meilisearch Instant Search + PostgreSQL ILIKE fallback
Forms & State	React Hook Form + Zod, TanStack Query v5, Framer Motion
DevOps & Hosting	PM2 Process Manager, GitHub Actions, Hostinger VPS, Cloudflare CDN

7. Key Results & Production Impact

<50ms Step Transitions
Valuation flow responds under 50ms per step using client-side preloaded cache.

Zero Downtime Migration
Migrated 49k+ records from agency database to Neon PostgreSQL seamlessly.